Accessibility of semantic content across discourse boundaries

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A growing body of work shows that information in subordinate discourse–typically parenthetical and non-at-issue–becomes less accessible from working memory (WM) after processing.[1-2] For example, wh-dependency processing costs are reduced when the embedded clause is part of subordinate discourse[1; 3-5]; subject-verb number agreement attraction effect disappears when the distractor is embedded in subordinate discourse[2; 6-7]. However, whether this reduced accessibility extends to *semantic* content remains less understood. **In this work, we ask:** *to what extent does semantic content in subordinate discourse remain accessible?* We hypothesize that, if the semantic content inside subordinate discourse becomes less accessible across discourse boundaries, the processing of the downstream discourse context should show reduced sensitivity to the semantic features encoded inside subordinate discourse. **Findings:** we provide new evidence that semantic content remains accessible across discourse boundaries, but that its degree of accessibility depends on whether the context has explicitly provided the referential domain (i.e., the set of possible candidates of discourse referents).

Design. We conducted two English reading experiments using the A-Maze paradigm[8] in a 2x2 design (RC x semantic relatedness) (see (1)). Following earlier work[1-2], we manipulated discourse status using appositive relative clauses (ARCs), which contribute subordinate and parenthetical discourse, and restrictive relative clauses (RRCs), as a stand-in for main discourse[9]. To test the accessibility of subordinate semantic content, we manipulated semantic relatedness (high vs. low) between the embedded noun in the RC (e.g., *professor*/*ballerina*) and the downstream main verb (*grade*). **Exp 1** (*N*=82; 32 items) and **Exp 2** (*N*=100; 32 items) were identical, except that **Exp 2** included a context establishing the referential domain with one or multiple discourse referents that support natural ARC or RRC readings. **Predictions.** As in [10-11], we expect a baseline semantic-relatedness effect at the main verb (*grade*), with longer reading times (RTs) in the low-condition due to reduced semantic overlap with the embedded noun. Crucially, if subordinate semantic content is less accessible in downstream main discourse, this semantic-relatedness effect should be reduced in ARCs relative to RRCs.

Results (Fig. 1). Bayesian linear mixed-effects models on RTs in the critical and spillover regions (Tab. 1) first show a robust baseline semantic-relatedness effect in both experiments, where RT_low > RT_high. We also found longer RTs in RRCs than ARCs in the spillover region in both experiments. Importantly, we identified a 2-way interaction in both experiments, but in opposite directions. In **Exp 1** (@spillover), the semantic effect was reliably reduced in ARCs, in line with our prediction. In contrast, in **Exp 2** (@critical), with a context sentence, this effect was reversed, such that the semantic effect was reliably larger in ARCs.

Discussion. The robust main effect of semantic relatedness shows that semantic content remains accessible across discourse boundaries. However, the interaction suggests that the *degree* of accessibility is modulated by discourse status. Importantly, this modulation depends on whether the referential domain is explicitly established in the context. While Exp 1 shows reduced accessibility of semantic content in subordinate ARCs at the main verb–consistent with prior work–Exp 2 exhibits the reverse: with an explicit referential domain, main-verb processing becomes more sensitive to semantic content in ARCs than in RRCs. No existing account straightforwardly predicts this reversal as far as we are aware of. We conjecture that when the referential domain is explicit (as in Exp 2), the parser is actively engaged in identifying the referent among multiple candidates while constructing the main proposition. In RRCs, where the RC is crucial for referent identification (e.g., which lady?), more resources are consumed in RC processing, leaving fewer available for integrating the downstream main verb and its semantic compatibility with prior context. In ARCs, by contrast, processing the subordinate discourse demands fewer resources as the referent is already uniquely identified, allowing greater sensitivity to ARC semantic content at the main verb. As such, the present findings highlight the dynamic interaction between the local and global discourse in sentence processing.

(1) **Example stimuli.** Target sentence: ARC condition (with commas) in (a); RRC condition in (b). Context sentences appeared only in Exp 2. Sentences were presented word-by-word in an A-Maze task (comma presented with the noun). Semantic relatedness between the embedded noun and the main verb (*grades*) was manipulated: high (e.g., *professor*) vs. low (e.g., *ballerina*). GloVe-based cosine similarities[12] were higher in the high- than in the low-condition for each item. Critical regions are bold-faced; regions of interest are italicized.

a. Context : Amy chatted with a lady last night. [only in Exp 2]
   Target : The lady, who is a {professor/ballerina}, routinely ***grades*** *philosophical* essays.
b. Context : Amy chatted with three ladies last night. [only in Exp 2]
   Target : The lady who is a {professor/ballerina} routinely ***grades*** *philosophical* essays.

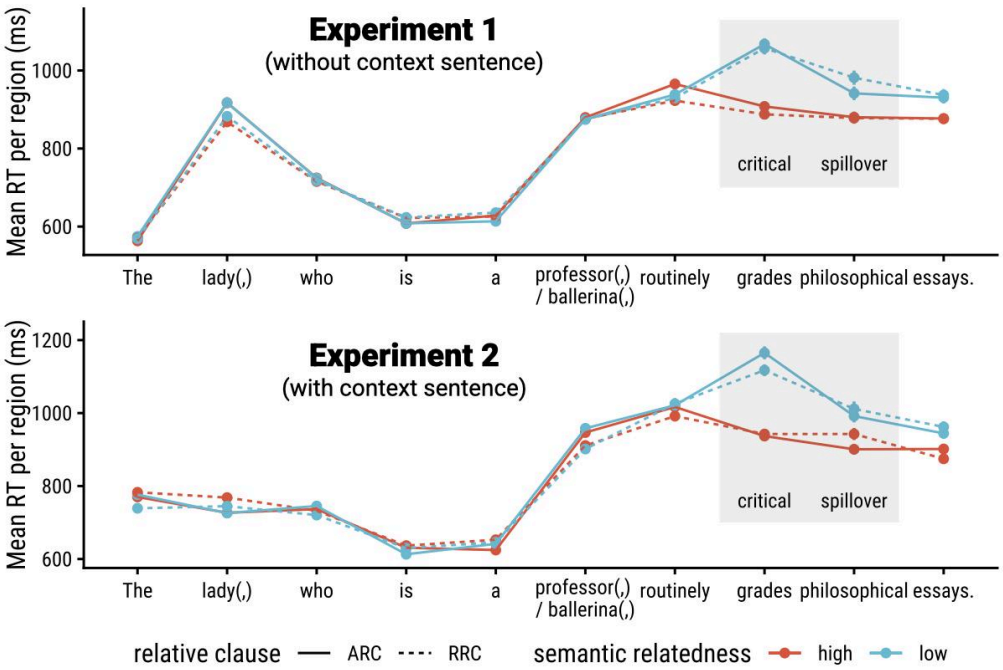

Figure 1. Region-by-region reading times of target sentences.

	Experiment 1				Experiment 2			
	Critical		Spillover		Critical		Spillover	
	b	p(|b|>0)	b	p(|b|>0)	b	p(|b|>0)	b	p(|b|>0)
Sem	**0.077**	**1**	**0.042**	**1**	**0.091**	**1**	**0.046**	**1**
RC	0.008	0.896	**-0.010**	**0.951**	0.008	0.899	**-0.009**	**0.909**
Sem:RC	0.000	0.534	**-0.013**	**0.971**	**0.009**	**0.905**	0.006	0.797

Table 1. Statistical results (*Sem*: semantic relatedness; *RC*: RC type).

References. [1] Dillon et al. (2017) *JML*. [2] Kim & Xiang (2024) *CogSci*. [3] Dillon et al. (2014) *LCN*. [4] Kroll & Wagers (2019) *manuscript,* UC Santa Cruz. [5] Duff et al. (2023) *Glossa Psycholinguistics.* [6] Ng & Husband (2017) *CUNY.* [7] McInnerney & Atkinson (2020) *CUNY.* [8] Boyce et al. (2020). *JML.* [9] Jasinskaja (2016) *manuscript,* Univ. of Cologne. [10] Pynte et al. (2008) *Vision Research*. [11] Levy (2008) *Cognition.* [12] Pennington et al. (2014). *EMNLP.*
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